# Trekking Management Application — Project Report

## 1. Student Details

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**Course:** Modern Application Development 1 - Project

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## 2. Project Details

### Project Title: Trekking Management Application

### Problem Statement:

Adventure organizations require efficient systems to manage trekking activities involving trek organizers, staff, and participants. Currently, many trekking groups rely on spreadsheets, phone calls, or manual coordination, which makes it difficult to manage trek approvals, track bookings, avoid overbooking, and maintain trek history.

This project builds a web-based Trekking Management Application that allows **Admin, Trek Staff, and Users (Trekkers)** to interact with the system based on their roles.

### Approach:

The application was built using Flask as the backend framework with session-based role authentication. The three roles — **Admin, Trek Staff, and Trekker** — each have separate dashboards with restricted access based on their role.

The database was designed using SQLAlchemy with five models: **Admin, Staff, User, Trek, and Booking**. All routes, business logic, and validations are handled server-side in Flask, with Jinja2 used to render dynamic HTML pages.

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## 3. AI/LLM Declaration

I used Claude Sonnet 4.6 for Flask route logic, SQLAlchemy model definitions, HTML templates and used ChatGpt (GPT - 5) for implementation related guidance and thinking of colour theme. The extent of AI/LLM usage is around **20–25%**. All final implementation logic, debugging, integration was carried out by me manually.

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# 4. Technologies and Frameworks Used

| Technology / Library | Purpose |
|---|---|
| Flask | Core backend web framework |
| SQLAlchemy | Object Relational Mapper for SQLite database |
| Jinja2 | Template engine for rendering dynamic HTML pages |
| Bootstrap 5 | Frontend styling and responsive design |
| Flask-WTF / WTForms | Form creation and validation |
| Werkzeug | Secure password hashing |
| SQLite | Lightweight local database |
| Google Fonts | Typography (Playfair Display, Inter) |

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# 5. Database Schema / ER Diagram

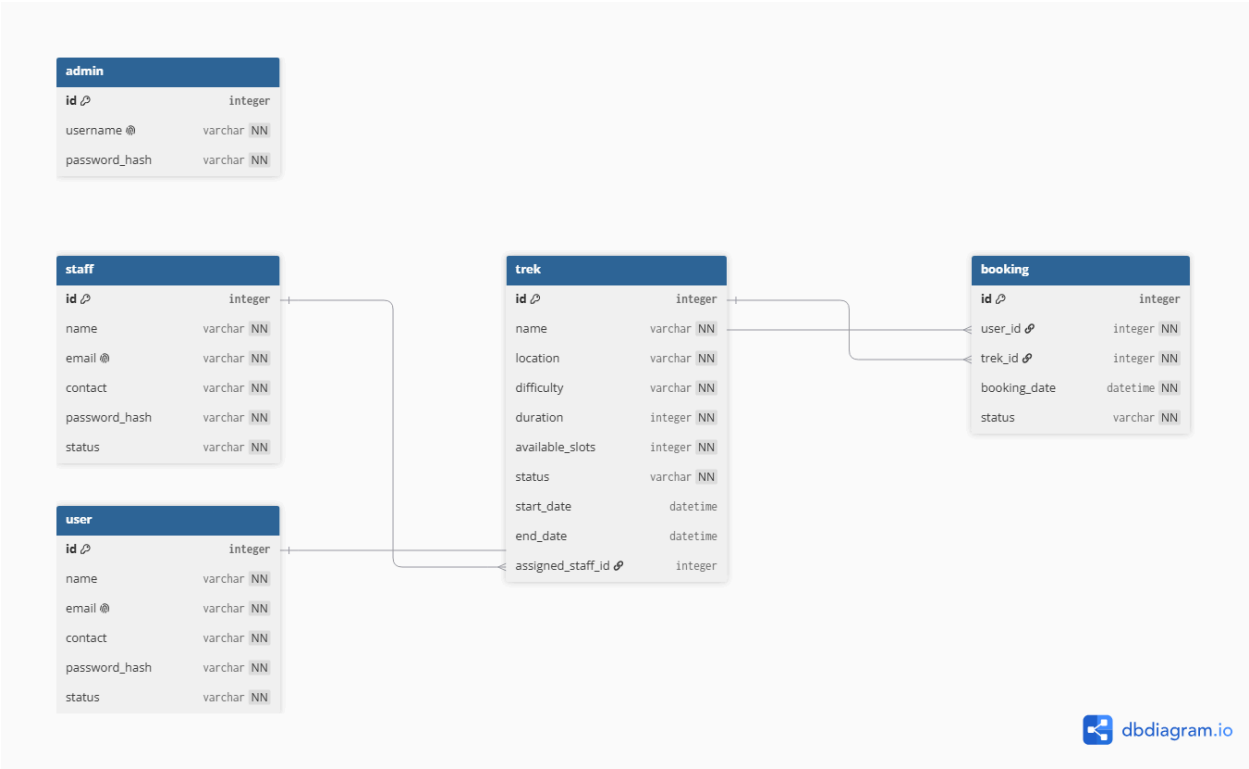
### Tables:

1. **Admin** — stores admin credentials (`id`, `username`, `password_hash`)
2. **Staff** — stores trek staff details (`id`, `name`, `email`, `contact`, `password_hash`, `status`)
3. **User** — stores trekker details (`id`, `name`, `email`, `contact`, `password_hash`, `status`)
4. **Trek** — stores trek information (`id`, `name`, `location`, `difficulty`, `duration`, `available_slots`, `status`, `start_date`, `end_date`, `assigned_staff_id`)
5. **Booking** — stores booking records (`id`, `user_id`, `trek_id`, `booking_date`, `status`)

### Relationships:

- One-to-Many → `Staff → Trek` (a staff member can be assigned to multiple treks)
- One-to-Many → `Trek → Booking` (a trek can have multiple bookings)

- One-to-Many → `User → Booking` (a user can have multiple bookings)




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# 6. Application Flow

## Admin

- Logs in with pre-existing credentials
- Can create, edit, and delete treks
- Can approve or blacklist trek staff
- Can assign staff to treks
- Can blacklist users
- Can view all bookings and search across users, staff, and treks

## Trek Staff

- Registers and waits for admin approval
- Once approved, can log in and view assigned treks
- Can update trek slots and status
- Can view participant list

## User (Trekker)

- Registers and logs in
- Can browse and search open treks
- Can book treks (overbooking and duplicate booking prevented)
- Can view booking status and trekking history
- Can edit their profile

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## 7. API Resource Endpoints

No separate REST API endpoints are implemented. The application uses Flask routes for handling requests and rendering the web pages.

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## 8. Core Business Rules Implemented

- Admin is pre-created in the database — no registration allowed
- Trek staff must be approved by admin before accessing dashboard
- Users can only book treks with status **"Open"**
- Overbooking is prevented — available slots decrease on each booking
- Duplicate bookings are not allowed
- Only assigned staff can manage a trek
- Blacklisted users and staff cannot log in

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## 9. Video Presentation

**Drive Link:** trekking management app demo vid 25f2006483.mp4
https://drive.google.com/file/d/16dEP9qk_KrXUhXIW0ru9ERLCCZasdBpf/view?usp=sharingv